COMPUTER SCIENCE & IT

DIGITAL LOGIC

Lecture No: 04

Miscellaneous Topics

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Recap of Previous Lecture

Number System

Topics to be Covered

- Number System Cont.
- PI, EPI

Question

Two numbers A and B are represented using 2's compliment representation :

$2's \to (0101)_2 = -A$

$$A = (1011)_2 = (-5)_{10} \qquad B = (1110)_2 = (-2)$$

$-(2^3)$ to $-(2^3-1)$

$\boxed{-8 \text{ to } 7}$

Then the result of (B – A) will be

(a) $(0011)_2$

(b) $(10011)_2$ ✗

(c) $(1000)_2$ ✗

(d) $(1101)_2$ ✗

$$\boxed{\begin{aligned} B-A &= -2-(-5) = (3)_{10} = (0011)_2 \\ A-B &= -5-(-2) = (-3)_{10} = (1101)_2 \end{aligned}}$$

$B - A = B + (-A)$

$$\begin{array}{r} 1110 \\ + \ 0101 \\ \hline 1\,0011 \end{array}$$

discard (1) ; $0011 \to$ Result $= (0011)_2$

[Question]

Three numbers A, B and C are represented using 2's compliment representation as :

$$A = (1010)_2, \quad B = (0111)_2, \quad C = (1100)_2$$

$= (-6)_{10}$ $\quad = (+7)$ $\quad = (-4)$

2's $\rightarrow (0100)_2 = (-C)$

$A + B + C = (-3)_{10} = (1101)_2$

Then the results of $(A + B - C)$ in 2's compliment representation as :

(a) $(0011)_2$

(b) $(0101)_2$ ✓

(c) $(1011)_2$

(d) $(1101)_2$

$A + B + (-C) = (5)_{10}$

$$\begin{array}{r} 1010 \\ +\ 0111 \\ \hline \cancel{1}\,0001 \end{array} \Rightarrow (A+B)$$

discard

$$\begin{array}{r} 0001 \\ 0100 \\ \hline 0101 \end{array}$$

$\rightarrow$ Result of $(A+B-C) = (0101)_2$

$A = 1011 = (-5)_{10}$

$B = 1100 = (-4)_{10}$

$C = 0011 = +3$

$-5-4+3 = -6 = 1010$

$(A+B+C)$

$\longrightarrow$

$$\begin{array}{r} 1011 \\ 1100 \\ \hline (1)\,0111 \end{array} \longrightarrow (A+B)$$

discard

$$\begin{array}{r} 0111 \\ 0011 \\ \hline 1010 \\ \hline \end{array} \longrightarrow \text{Result}$$

Question

A signed no. is represented using 2's compliment representation as :

$$N_1 = (1001)_2 = (1111\,1001)_2$$

Then its 8-bit representation will be :

(a) $(1001\,1001)_2$

(b) $(1111\,1001)_2$

(c) $(1000\,1001)_2$

(d) $(0000\,1001)_2$

Question

A 4-bit number in 2's compliment sign no. representation is

$M = \boxed{S_3}\boxed{S_2}\boxed{S_1}\boxed{S_0}$ $= (0001)_2 = (+1)_{10}$

and other no. given in 2's compliment representation is

$N = \boxed{S_3}\boxed{S_3}\boxed{S_2}\boxed{S_1}\boxed{S_0}\boxed{1}\boxed{0}$

Then the relation between M and N is :

(a) $N = 4M$ ✗

(b) $M = 4N + 2$ ✗

(c) $N = 4M + 2$ ✓

(d) $N = 4M + 1$ ✗

$N = 0000110 = (+6)_{10}$

$M = S_3 S_2 S_1 S_0$

$M = S_3 S_3 S_2 S_1 S_0$

$4M = S_3 S_3 S_2 S_1 S_0 0 0$

$+\ 2 \quad 0000010$

$N = S_3 S_3 S_2 S_1 S_0 1 0$ Result

$(-17)_{10} \longrightarrow (101111)_2$

$(+9)_{10} \longrightarrow (01001)_2$

$(010001)_2$

$\Big\downarrow 2's$

$(101111)_2$

$-(2^{n-1})$ to $(2^{n-1}-1)$

$(-43)_{10} \longrightarrow (1010101)_2$

$\longrightarrow (0101011)_2 \xrightarrow{2's} (1010101)_2 = -43$

$\downarrow$

$(11111\,1010101)_2 = -43$

BCD Codes

8421 code

Weighted code system

Non-self complimentry

Excess-3 code → unweighted code.

Excess-3 code → self complimentry code

Decimal	BCD	Excess-3 code
0 →	0000	0011
1 →	0001	0100
2 →	0010	0101
3 →	0011	0110
4 →	0100	0111
5 →	0101	1000
6 →	0110	1001
7 →	0111	1010
8 →	1000	1011
9 →	1001	1100

1010
1011
1100
1101
1110
1111

→ invalid BCD codes.

0000
0001
0010
1101
1110
1111

→ invalid Excess-3 code.

$$25 = (00100101)_{BCD}$$

$$= (01011000)_{Excess-3}$$

Note: → Excess-3 code is the only weighted code system which is self-complimentry in nature.

- In case of weighted code system if summation of weights is equal to 9 then it will be self complimentry in nature.

  eg. 2421, 3321, 4311 etc.

Gray Code / Number System

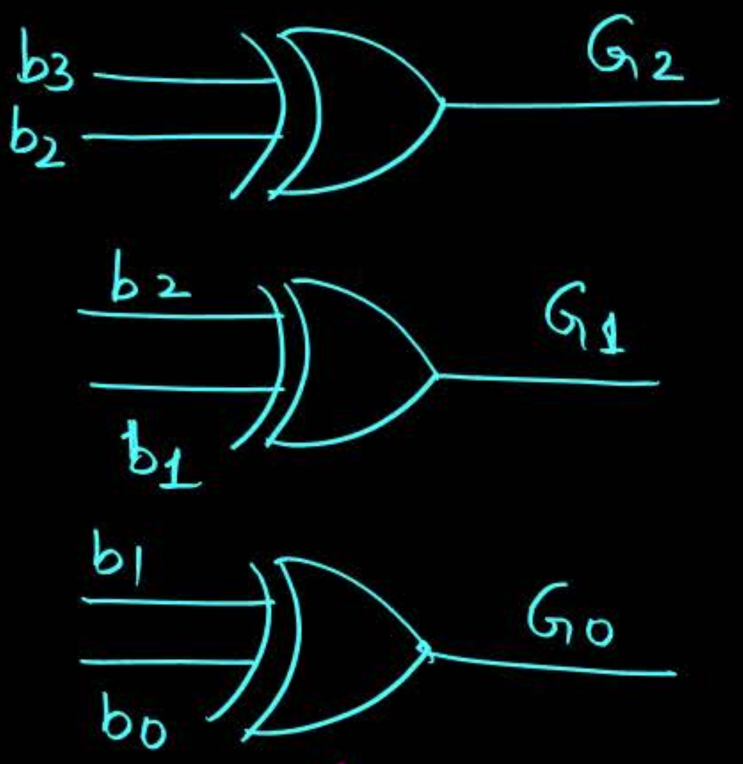
- Unweighted Numbu system
- UDC → unit distance code

	Binary	Grey
24	11000	10100
25	11001	10101
26	11010	10111
27	11011	10110
28	11100	10010
29	11101	10011

Binary to Gray Code Conversion :

$$\overset{B}{b_3 b_2 b_1 b_0} \longrightarrow \overset{G}{G_3 G_2 G_1 G_0}$$

$$b_3 = G_3$$



$(121) \longrightarrow (1000101)_G$

$\quad \longrightarrow (1111001)_B$

$$\boxed{B \text{ to } G}$$

B	G
101010	111111

$(11011111)_G$

$\downarrow$

$(10010101)_B$

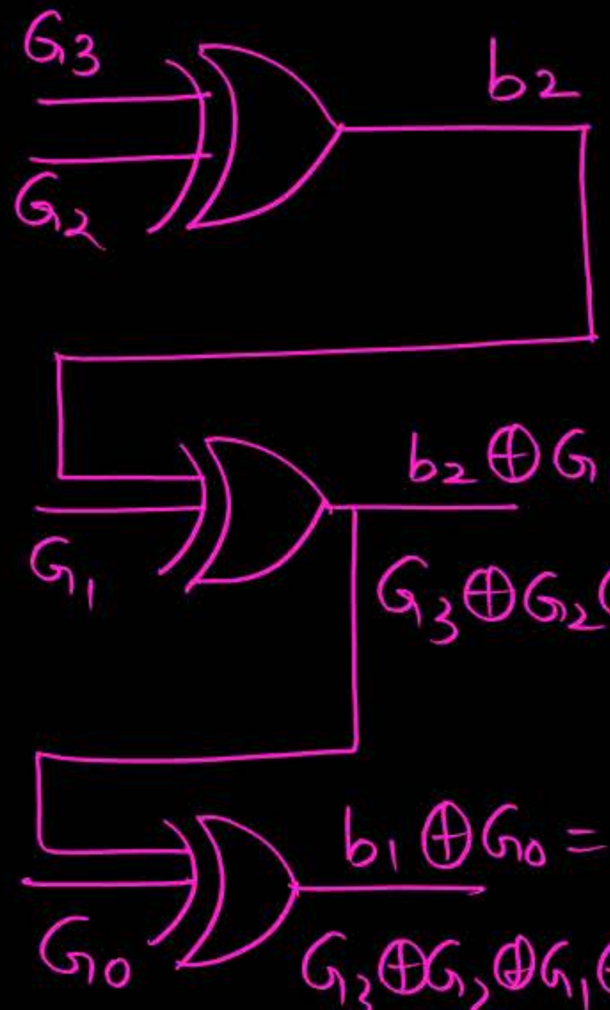
• $(43)_{10} \longrightarrow \overset{B}{(101011)_2} \longrightarrow \overset{G}{(111110)}$

	B	G
• $(121)_{10} \longrightarrow$	$(1111001)_2$	(1000101)
$(149)_{10} \longrightarrow$	$(10010101)_2$	(11011111)
$(78)_{10} \longrightarrow$	$(1001110)_2$	(1101001)

Gray to Binary Conversion :

$G_3 G_2 G_1 G_0 \qquad b_3\ b_2\ b_1\ b_0$

$G_3 = b_3$



$b_2 \oplus G_1 = b_1$

$G_3 \oplus G_2 \oplus G_1 = b_1$

$b_1 \oplus G_0 = b_0$

$G_3 \oplus G_2 \oplus G_1 \oplus G_0 = b_0$

Implicants

All possible 1's, all possible pairs, all possible quads, all possible groups are called as implicants.

PI → The implicant [group] which can't be combined with another implicant [group] to convert into bigger group is called as PI.

EPI: The PIs having atleast single '1' which is covered only by that PI then this PI is called as EPI.
If all the '1's of a PI can be covered other PIs also then that PI is not a essential PI.

	$\overline{B}$	B
$\overline{A}$	1	1
A	1	

Implicants $\overline{A}\,\overline{B}$, $\overline{A}B$, $A\overline{B}$

$\overline{B}$, $\overline{A}$

PI → $\overline{B}$, $\overline{A}$

	$\overline{B}\,\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$		1	1	
A		1	1	1

Implicants : 11

PI → C, AB

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	1	1		
A			1	1

$PI \longrightarrow \bar{A}\bar{B},\ AB$

$EPI \longrightarrow \bar{A}\bar{B},\ AB$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	1	1		
A		1	1	

$PI \longrightarrow \bar{A}\bar{B},\ AC,\ \bar{B}C$

$EPI \longrightarrow \bar{A}\bar{B},\ AC,$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	1	1	1	
A			1	1

PI $\longrightarrow$ $\bar{A}\bar{B}$, AB, $\bar{A}C$, BC

EPI $\longrightarrow$ $\bar{A}\bar{B}$, AB

Examples :

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$		1		
$\bar{A}B$		1	1	1
AB	1	1	1	
$A\bar{B}$			1	

Prime Implicants $\rightarrow$ BD, $\bar{A}\bar{C}D$, ACD, $\bar{A}BC$, $AB\bar{C}$

Essential Prime Implicants $\rightarrow$ $\bar{A}\bar{C}D$, ACD, $\bar{A}BC$, $AB\bar{C}$

Non-Essential Prime Implicants $\rightarrow$ BD

Examples :

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1		
$\bar{A}B$	1	1	1	
AB		1	1	
$A\bar{B}$	1	1		

Prime Implicants → $BD, \bar{B}\bar{C}, \bar{A}\bar{C}, \bar{C}D$

Essential Prime Implicants → $BD, \bar{B}\bar{C}, \bar{A}\bar{C}$

Non-Essential Prime Implicants → $\bar{C}D$

Topic : 2 Min Summary

Numbu system
PI, EPI.

Thank you

GW

Soldiers !